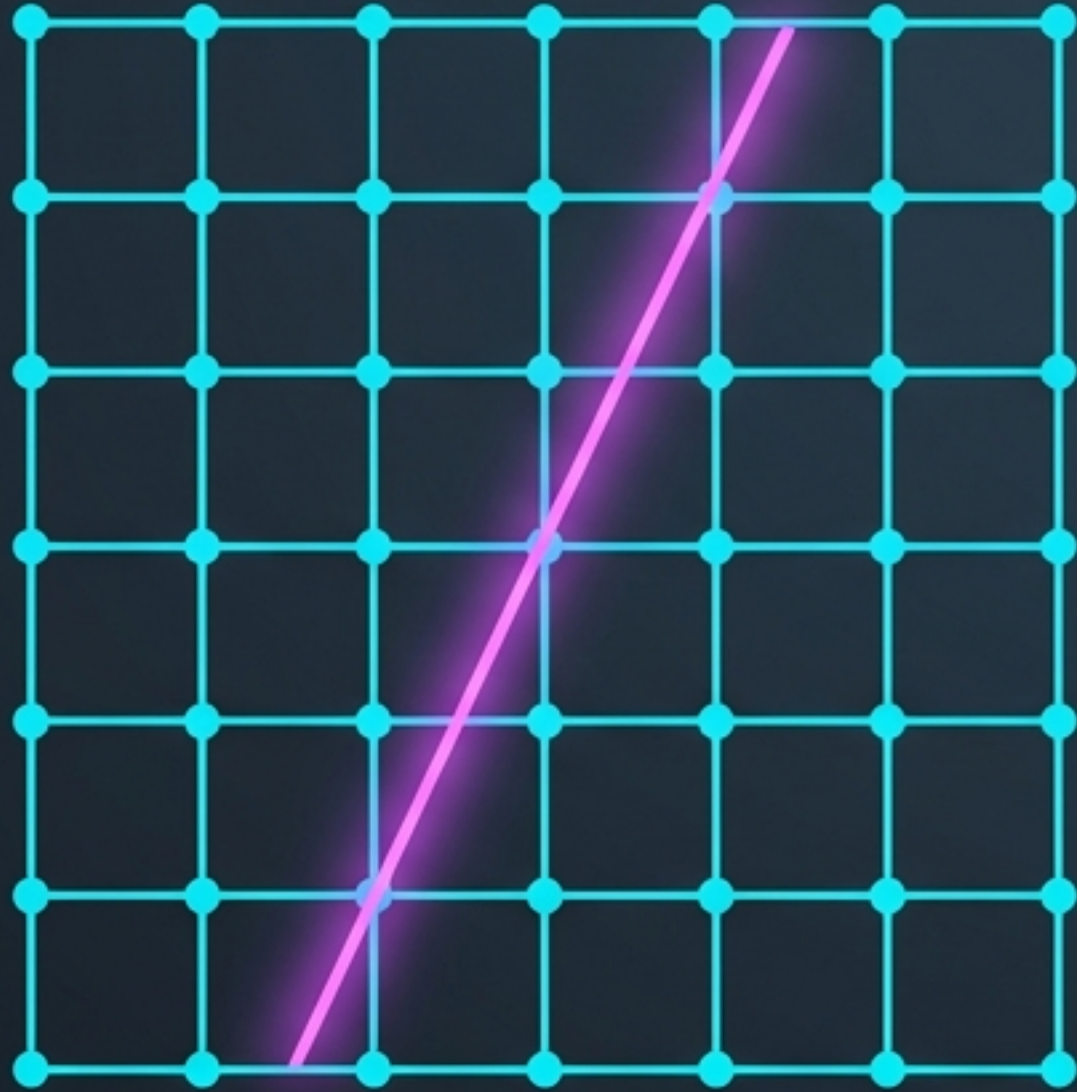


$$\left(\frac{3x^3 - 7x^2 + 2x}{x - 2} \right) = \begin{array}{c} 3x^2 - x \\ \vdots \\ \{ \dots, 0, 2, 10, \dots \} \end{array}$$

The End of a 126-Year Mystery

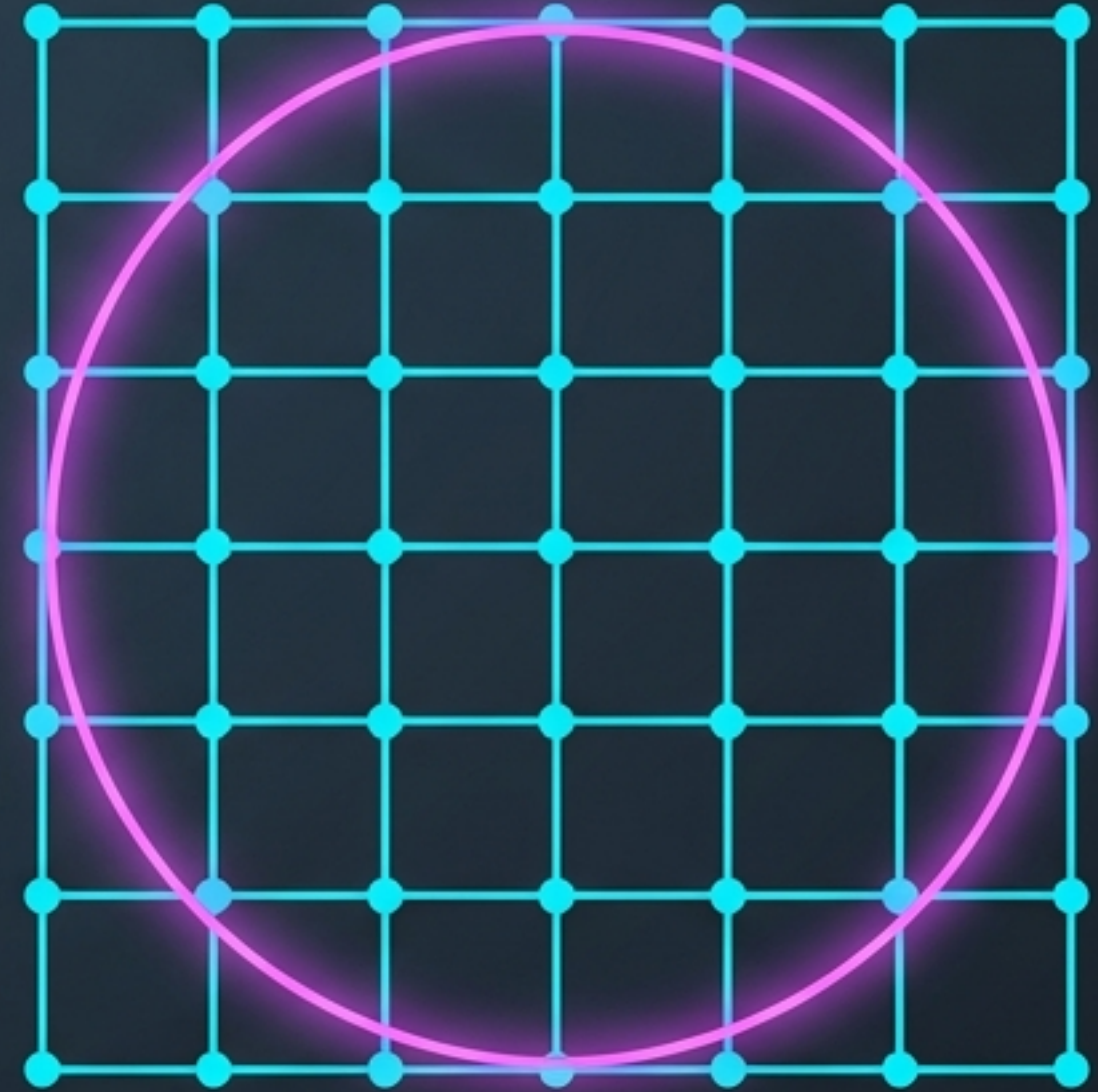
How Artificial Intelligence Settled Hilbert's Tenth Problem for the Rationals

$$2x = 1$$



Has a rational solution ($x = 1/2$)
but NO integer solution.

$$x^2 + y^2 = 3$$



Has NO rational or integer solutions.

A Diophantine equation is simply a polynomial where we are only allowed to ask one question: Are there solutions inside a specific mathematical universe?

The Flashlight Problem

Searching confirms presence, but never absence. To truly solve Hilbert's problem, an algorithm must be able to confidently stop and say NO. In logic, an inability to guarantee a stop is known as the Halting Problem.

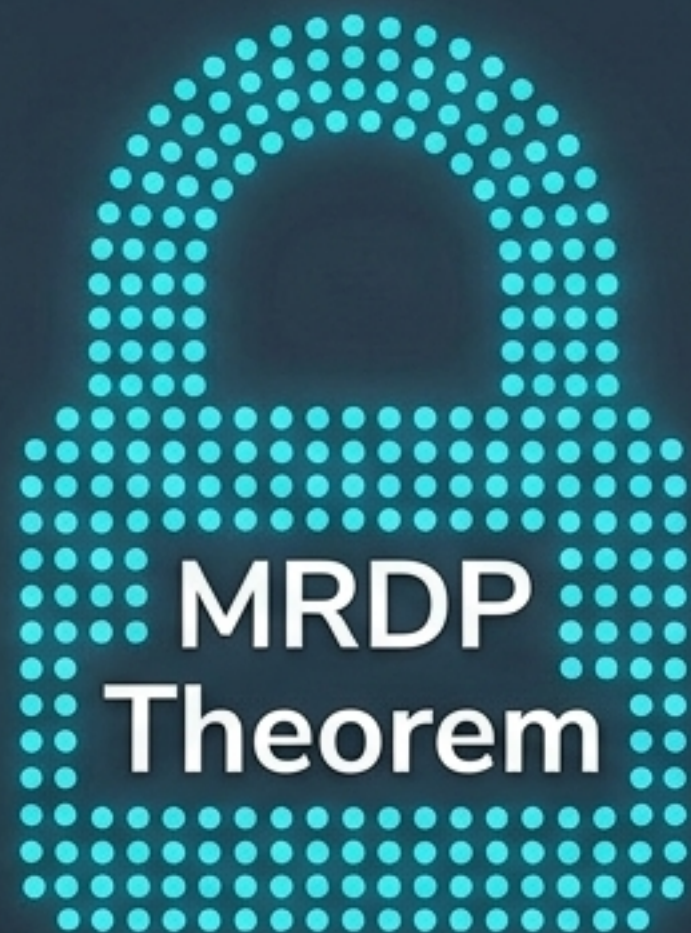


If a solution exists, searching
1, 2, 3... will eventually find it.



If a solution does
NOT exist, you
search to infinity.





1900: David Hilbert
poses his 10th Problem.



1961: Davis, Putnam,
and Robinson (DPR)
reduce the problem.

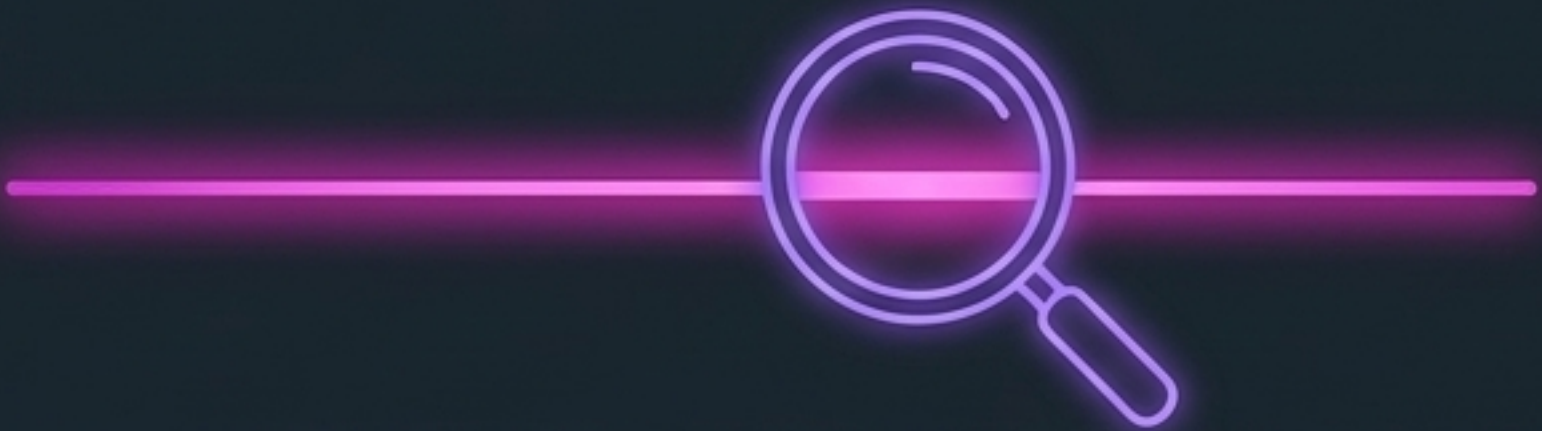


1970: Yuri Matiyasevich
delivers the final blow.



The Integers ($\mathbb{Z}$) are fundamentally undecidable. No algorithm can ever be built that always tells you if an integer solution exists. The first half of Hilbert's dream is dead.

The $\mathbb{Z}$ vs $\mathbb{Q}$ Matrix

Integers $\mathbb{Z}$	Geometry	Rationals $\mathbb{Q}$
		
The Logic Trap		
Diophantine equations only allow “There Exists” statements.		Previous attempts to capture integers inside rationals (Robinson 1949, Poonen 2009, Koenigsmann 2016) all required “For All” statements, breaking the rules of the game.

Mazur’s Conjecture suggests that the smooth, dense geometry of rationals makes a simple, rule-abiding definition of integers geometrically impossible.

Corollary 7.1

rational solvability has Turing degree $0'$, the degree of the halting problem.

Theorem 1.1:

There is no algorithm that decides whether a polynomial with integer coefficients has a rational zero.

Corollary 7.1 (i)

polynomials of degree at most 4 are already as hard as the general problem.

The Logic Race: Integers vs. Rationals

Program 1: Search for an Integer Solution.



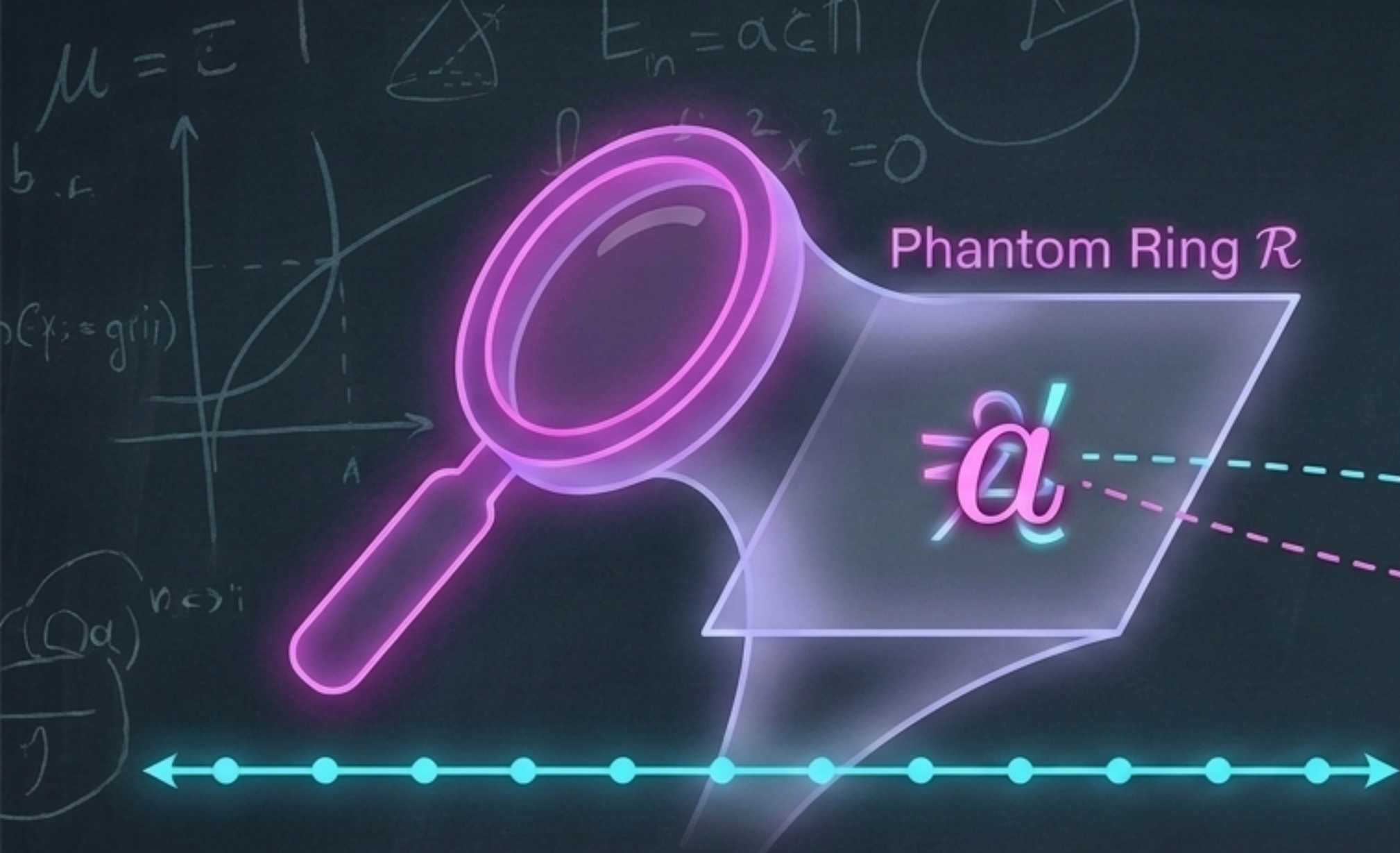
Program 1: Search for an Integer Solution.

IF AND ONLY IF

Program 2: Run Finite Rational Tests searching for a Failure.

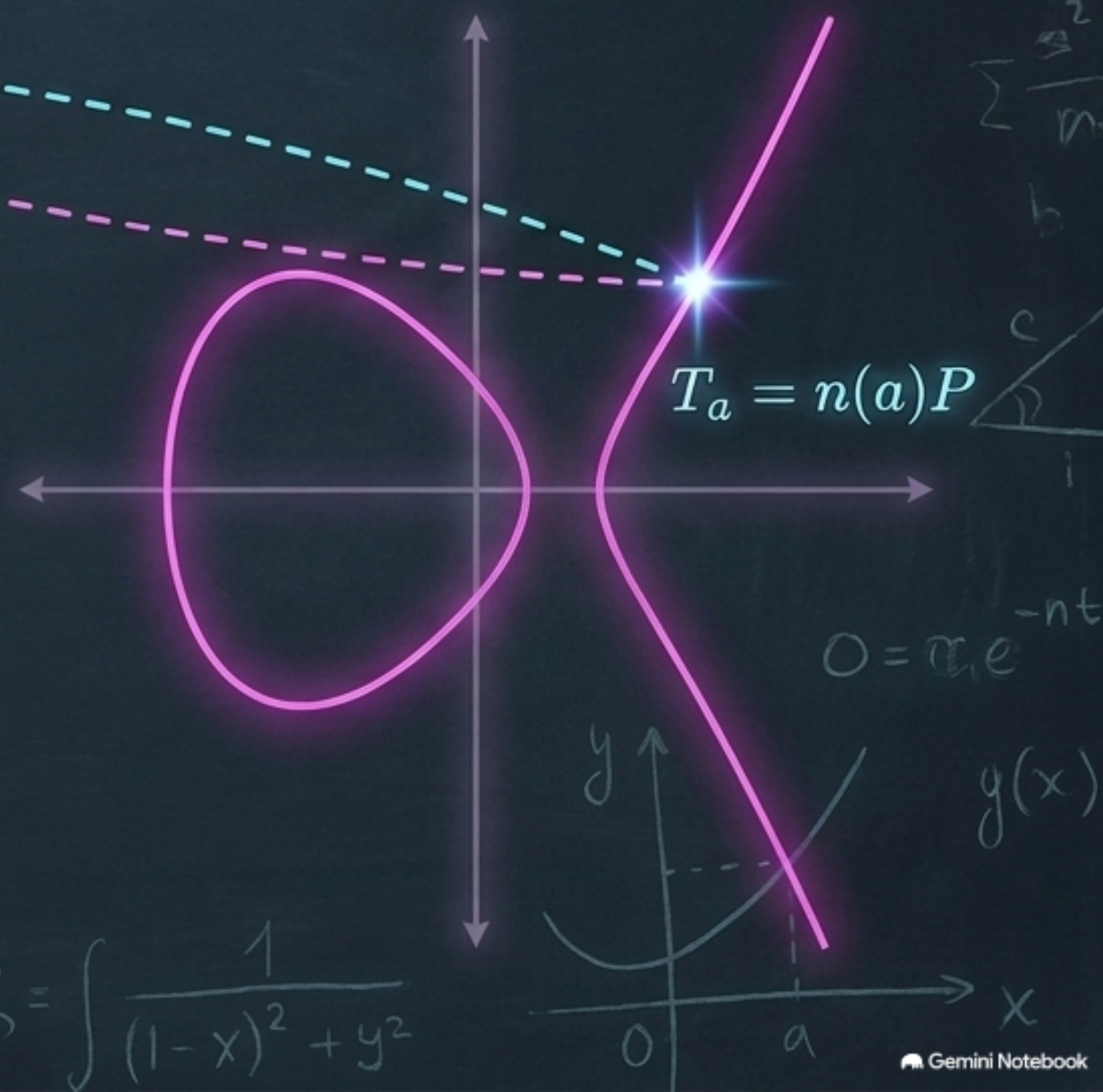
Program 2: Run Finite Rational Tests searching for a Failure.

The master trap: An integer solution exists if and only if every single rational test succeeds. Therefore, one of these programs MUST eventually stop. If we had an algorithm for Q , we could run this race and solve the integer problem—but the 1970 MRDP theorem proved integers are unsolvable! Contradiction achieved.



$$y^2 = x^3 - d^2x$$

over $\mathbb{Q}(\sqrt{2})$, with $d = 75 - 53\sqrt{2}$



If every finite rational test succeeds, the compactness theorem of logic gives a ring R containing a root a of f . Each element a gets a point $T_a = \eta(a)P$ on the rank-one curve $y^2 = x^3 - d^2x$ over $\mathbb{Q}(\sqrt{2})$, with $d = 75 - 53\sqrt{2}$. The map η respects addition; it need not respect multiplication.

$$R = \int \frac{1}{(1-x)^2 + y^2}$$

The Prime Microscope

x

$$a = n(a) \bmod q^k$$

Original
Elliptic
Curve

Conjugate
Twin

Contact Primes (q)

To fix the multiplication problem, we examine the fractions $\frac{\text{prime}}{\text{prime}}$ by prime.

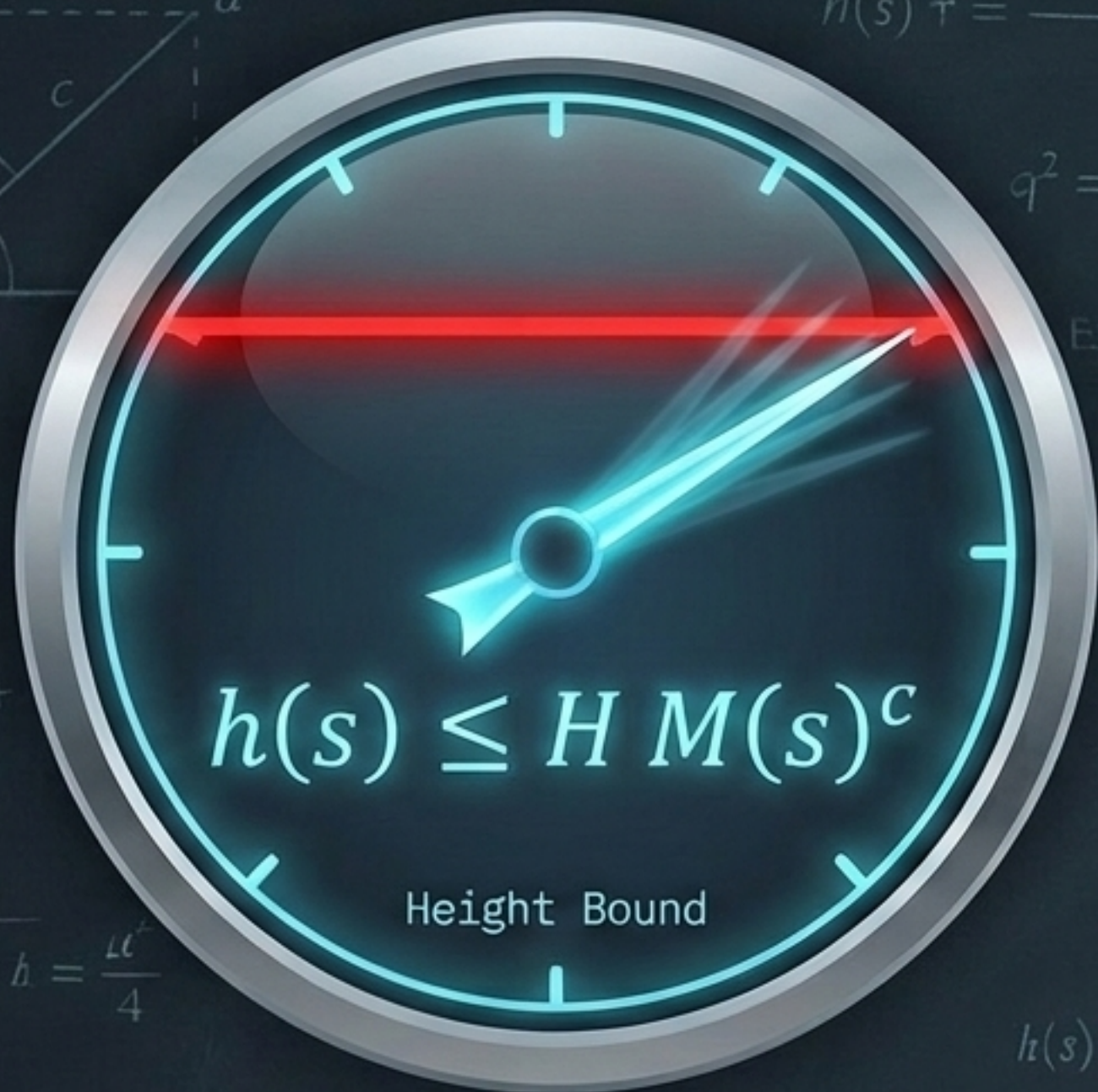
Using a "Truncated Formal Logarithm" on both the curve and its twin, we force the phantom elements to mathematically match up with reality to a massive degree of precision (K).



$$h(s) \tau = \frac{zx^2 - q\tau^2}{x}$$

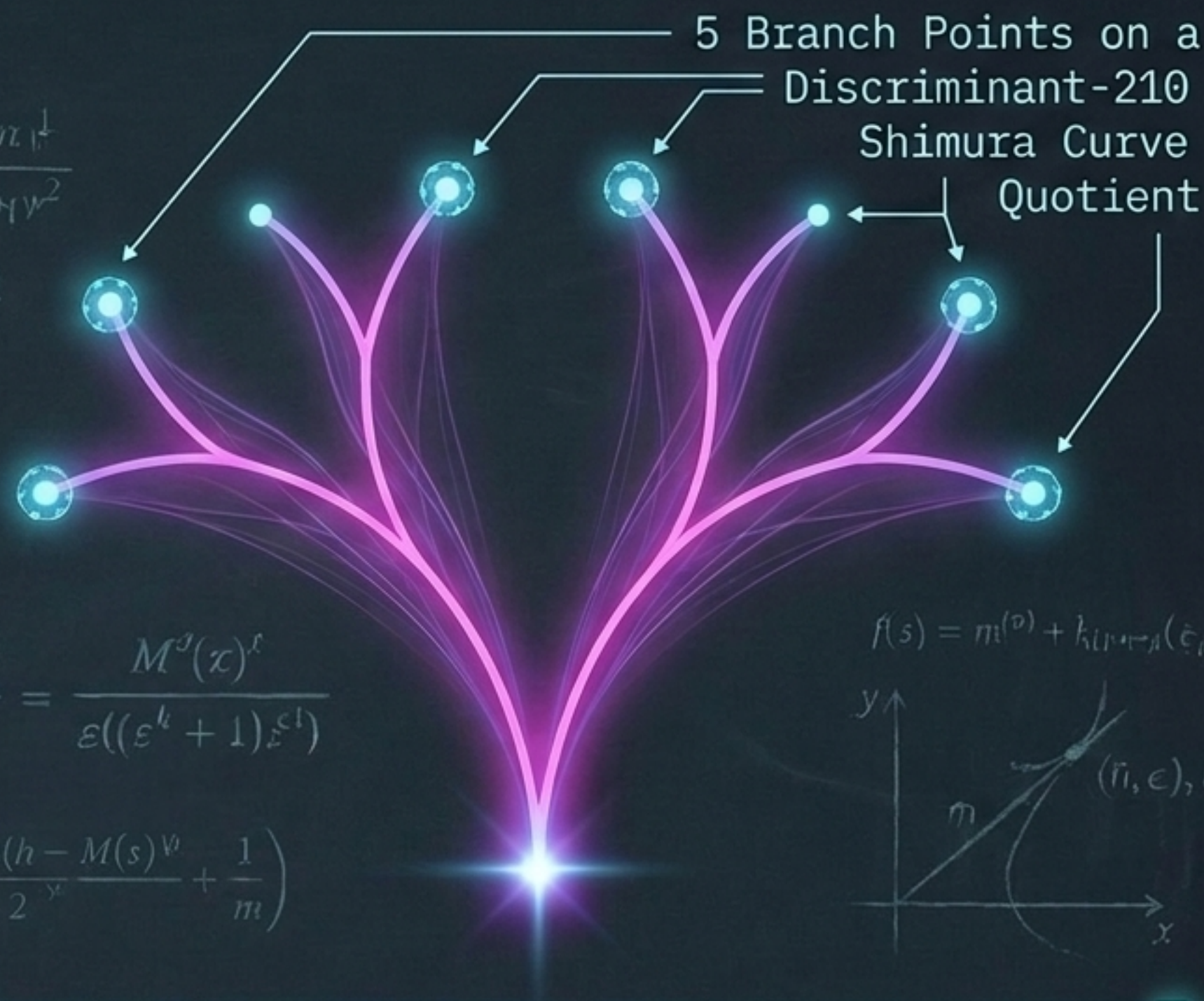
$$q^2 = \frac{n_a}{4n} = \frac{n_1^2}{4n^2}$$

$$\text{Exc}, \sigma = 2$$



$$h(s) \leq H M(s)^c$$

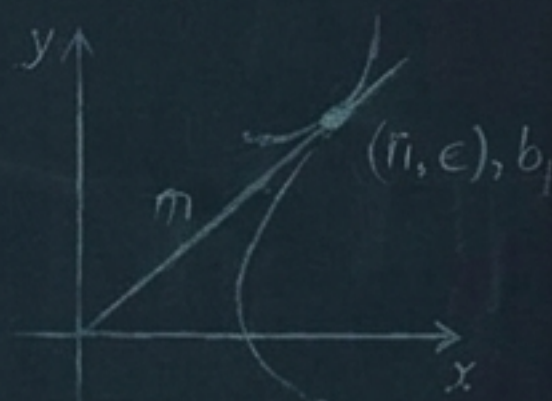
Height Bound



$$\frac{31 \varepsilon^2}{2} = \frac{M^2(x)^2}{\varepsilon((\varepsilon^k + 1)\varepsilon^l)}$$

$$h(s) = \left(\frac{1}{a_1 + 2} \frac{h(h - M(s)^2)}{x} + \frac{1}{m} \right)$$

$$f(s) = m^{(p)} + k_{(1, \dots, p)}(\tilde{c}_p)$$



For a fraction s outside the five points, $M(s)$ is the product of the primes at which s meets one of the five points to odd order. The height estimate $h(s) \leq H \cdot M(s)^c$ holds for every such s , so a fraction can be complicated only if it has odd contact at large primes.

The Parity Sieve

The pole-parity formula ϕ is true for every integer and forces denominators to have even exponents outside finitely many primes, by an elementary valuation argument (Lemma 3.1). The companion paper's pointwise 2-converse is used only to show that every integer satisfies ϕ (Lemma 3.4). ϕ lets each contact prime be matched with a tested branch (Lemma 2.10).

$$P = at^2[1-a, b, c]$$

$$\phi\pi = -\ln^n Z + \left(\frac{r+1}{2}\right)^2$$

$$\bar{V}_i = \left(\frac{1}{z+1}, \dots\right) - \left(\frac{\tau-1}{2}\right)$$

$$= \frac{1}{\sqrt{\pi+3}} \left(\frac{x+b}{2}\right)^2 [-c]$$

$$L_u = w|z|_b + \frac{1}{z^n} \left(\frac{1}{2} \left|1 - 2\ln \frac{1}{(n-b)}\right|^2\right)$$

$$h^2 = h^2 + bh$$

Rational
Possibilities

Height Bounds

C * B

Elliptic Point Growth

$$2 * \hat{h}(P) * B^2$$

$$\text{For } (x) = e\left(\frac{1}{2\pi}\right) (-b_1 x + 2)$$

$$= 2 \cdot \left(\frac{1}{2\pi}\right) (-2(x(P)^2))$$

$$= +^n \sqrt{\left(1 + \frac{1}{\sqrt{11x^2}}\right)}$$

$$\Gamma_{\text{psul}} = \sqrt{21 - \frac{1}{2\pi}}$$

The Squeeze.

A linear bound cannot contain quadratic growth forever.
The value B is crushed into a tiny, finite box. The
phantom rational root is squeezed until it has no
choice but to be a perfectly whole, discrete Integer.
The trap snaps shut. Contradiction.

$$2 * \hat{h}(P) * B^2$$

The Unopened Doors

No Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$ was actually found.

No “Many-One” Reduction.
(The logic map is Turing, not Many-One).

Mazur’s topology conjecture remains completely unresolved.

No Variable Bound.
(We don’t know the maximum number of variables needed).



The Author

Generated by Artificial Intelligence (OpenAI, Sept 2026).

The Status

Currently a preprint. It awaits rigorous human peer review or **machine formalization** (such as **Lean**) before being permanently etched into mathematical law.

A new era where machines solve the problems that human minds have hunted for a century.

The Ultimate Takeaway:

Deciding whether an equation has a rational solution is exactly as hard as deciding whether it has an integer solution: both are undecidable, with Turing degree 0prime.

Undecidable over $\mathbb{Q}$

1900: Hilbert's Challenge

1970: The Integer Defeat (MRDP)

2026: The Parallel Trap & The Elliptic Squeeze